

Social Event Prediction via Fourier Graph Learning

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Abstract

Social event prediction has garnered increasing attention in web-centered society. Most existing studies represent web-based event stream as chronological graph sequences, then leverage RNNs and GNNs to model temporal and relational patterns. However, this paradigm is inherently flawed: (1) RNNs struggle to capture long-term temporal dependencies, ignoring those temporally distant but influential events. (2) Spatio-temporal GNNs exhibit high computational complexity on large-scale real-time event streams, which hinders their web applications. To this end, we explore a novel paradigm called *Fourier Graph Learning* from the perspective of frequency domain. Specifically, we first define a novel data structure called Fourier Graph (FG). In FG, both nodes and edges are complex vectors, with real part encoding semantics and imaginary part representing semantic-specific temporal patterns. These temporal patterns are obtained by semantic-aware frequency filter, which utilizes semantics as guidance to adaptively incorporates both long-term dependency and short-term dynamic. Based on FG, we further propose Fourier Graph Neural Network (FGNN). It replaces time-domain convolution with frequency-domain multiplication for efficient aggregation. FGNN also includes a complex-valued event decoder, which fully leverages semantics and temporal patterns from complex space to predict future event probabilities. Extensive experiments show our superior performance with higher accuracy, less complexity and better interpretability compared with baselines.

CCS Concepts

• Information systems → Web mining.

Keywords

Social event prediction; Fourier analysis; graph neural network

ACM Reference Format:

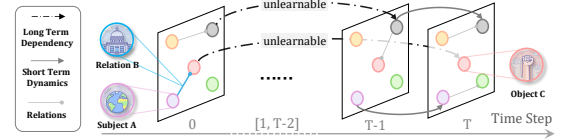
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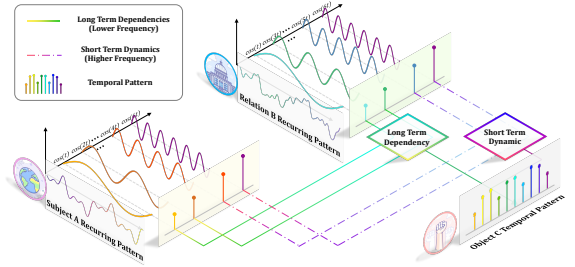


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(a) Existing paradigm on web-based event streams



(b) Fourier Graph Learning (FGL) paradigm

Figure 1: Illustration of our motivations: (a) Existing paradigm iterates on graph sequence in time domain to capture long-term dependency, while (b) FGL relies on frequency domain decomposition of event recurring patterns.

1 Introduction

Social event prediction, including the forecasting of events such as international conflicts [23] and public health [17], represents an important component of computational social science. As the Web has become the primary information source of propagation in recent years, predicting social events from web-based event streams has emerged as a prominent research hotspot. Reliable prediction of future social events is essential for modern society.

The task of social event prediction focuses on modeling the intricate temporal and relational patterns within web-based event stream. To achieve this goal, one popular formulation is to view each event in the stream as quadruple, i.e. (s, r, o, t) , where the symbols refer to subject, relation, object and timestamp, respectively. At each timestamp, these quadruples are grouped into event graph, ultimately forming a chronological sequence of event graphs. Based on this paradigm, numerous studies have emerged, which utilized various combinations of RNNs (e.g. LSTM and GRU) and GNNs (e.g. RE-GCN [11], SeCoGD [13], DHyper [29]) to model temporal and relational patterns respectively.

Despite the achieved progress, current paradigm still suffers from inherent limitations. Firstly, RNNs face challenges in **retaining**

long-term temporal dependencies, leading to ignorance of social events with delayed or long-lasting impacts. In real-world environments, long-term temporal dependencies are largely shaped by the recurrence of events (e.g. protests caused by political election campaigns, violent incidents at religious festivals), as evidenced by prior researches [15, 16, 38]. As shown in Figure 1(a), these recurrences often span very long time intervals, making it more difficult for RNNs to effectively capture them. Moreover, existing models rely on spatio-temporal convolutions to aggregate information, which pose **high computational complexity on large-scale real-time event streams**. In web-based scenarios, social event prediction often involves a vast amount of real-time events. Such complexity leads to excessive computational consumption and extended inference time, hindering their practical application for the Web.

To this end, we focus on solving the aforementioned issues from the view of frequency domain. As shown in Figure 1(b), frequency domain modeling focuses on discovering underlying periodicity of event recurring patterns, offering an alternative way to model long-term dependencies. Moreover, the Convolution Theorem enables the transformation from time-domain spatio-temporal convolution to frequency-domain multiplication, which largely reduces the computational costs. Based on these insights, we introduce a novel paradigm called *Fourier Graph Learning* (FGL). The core of FGL is a novel data structure called *Fourier Graph* (FG), whose nodes correspond to entities and edges represent events that occur between these entities (e.g. New Zealand express intent to cooperate with Greece, where "New Zealand" and "Greece" serve as nodes and "cooperate" serves as edge). In FG, both nodes and edges are complex vectors, with **real part encoding semantics and imaginary part representing semantic-specific temporal patterns**. These temporal patterns are obtained via semantic-aware frequency filters, whose parameters are determined by the semantics of the corresponding nodes and edges. Through this unique design, the semantic-aware frequency filters could adaptively select long-term dependencies (filtered low frequency components) and short-term dynamics (filtered high frequency components). In this manner, we directly encode these information into FG representations and facilitate explicit learning of long-term temporal dependencies. Based on FG, we further propose *Fourier Graph Neural Network* (FGNN). Specifically, instead of applying costly time-domain spatio-temporal convolution, **we utilize lightweight multiplication in the frequency domain as an alternative**. Moreover, FGNN also includes a complex-valued event decoder, which contains a feature reconstructor and a novel scoring function to decode multiplied complex vectors into probabilities of future events. This comes with two advantages: On one hand, frequency-domain multiplication facilitates direct interaction between temporal patterns and semantics, enabling deeper understanding of complicated events. On the other hand, by replacing both temporal convolutions of RNNs and spatial convolutions of GNNs, the computational complexity is significantly reduced. Our contributions are summarized as follows:

- We introduce *Fourier Graph Learning* (FGL), an innovative frequency-domain paradigm for social event prediction.
- We propose the *Fourier Graph* (FG), a novel data structure that utilizes complex-valued graph representations to explicitly model both semantics and temporal patterns.

- Based on FG, we propose the *Fourier Graph Neural Network* (FGNN), which leverages efficient frequency-domain multiplication to replace time-domain spatio-temporal convolutions, significantly reducing computational complexity.
- We verify the effectiveness of FGL on real-world web-based dataset, and results exhibit its superior performance with higher accuracy, less complexity and better interpretability.

2 Related Works

2.1 Social Event Prediction

As the Web has become one of the most significant information sources in recent years, web-based social event tasks have garnered growing interests, such as classification [19], detection [21] and prediction [20, 30]. Existing social event prediction methods are largely based on web-based event stream, which can be divided into four main paradigms: static graph based, temporal graph based, hypergraph based, and LLM based methods. (1) Static graph based methods view historical events as a static knowledge graph, and further apply 2-dimensional convolution [5], translational embedding [24], and relation-specific transformations [22]. However, these methods fundamentally fail to capture temporal dynamics and evolutionary processes inherent in historical events. (2) Temporal graph based methods define historical timeline as temporal knowledge graphs. To model both temporal and relational patterns, they usually combine both RNNs and GNNs [4, 11, 12], which face challenges in handling the long-term temporal dependencies and high computational complexity. (3) Hypergraph based methods are proposed based on temporal graph based methods. To solve the long-term temporal dependency challenges, these methods [13, 29] utilize hyper-edges to connect related entities in different timestamps. However, they typically increase computational complexity and pose new challenges in learning these high-order hyper-edges. (4) LLM based methods formulate the task by the paradigm of natural language generation, utilizing large language models as predictors. There are several works leveraging text data [25, 37] and multi-modal data [10]. However, they often rely on statistical patterns to generate responses rather than leveraging explicit knowledge, which frequently results in lack of factuality and causality.

2.2 Frequency Domain Learning

Recently, there's a growing trend of adopting frequency domain learning in many downstream tasks, such as time series forecasting [2, 34], spatio-temporal forecasting [33] and time series reconstruction [31]. Specifically, SFM [36] utilizes the Discrete Fourier Transform (DFT) to decompose the hidden state of the LSTM into its constituent frequency components. LCR [1] introduces a novel circular Laplacian kernel for temporal regularization to capture local trends and a circulation matrix nuclear norm for global trends. In contrast to prior approaches that employs frequency domain techniques to enhance existing architectures, we propose a novel paradigm that fully utilizes the complex-valued representations.

3 Preliminaries

In this section, we formulate the task and briefly introduce several related frequency-domain theorems. The used notations and descriptions are presented in **Appendix A**.

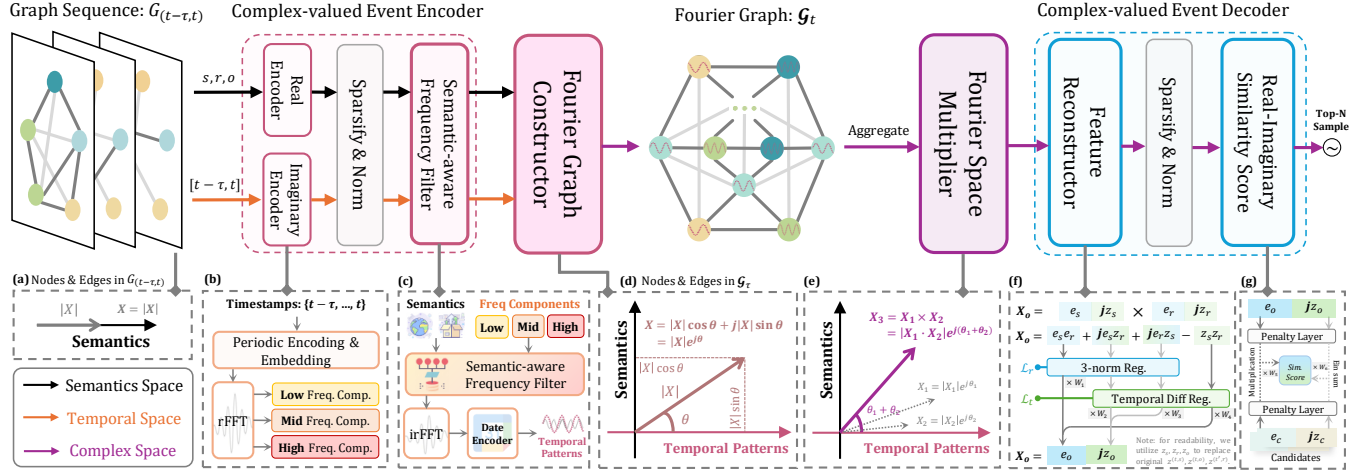


Figure 2: Workflow of our proposed *Fourier Graph Learning (FGL)*, a novel social event prediction paradigm that can be divided into two main stages: *Fourier Graph (FG) construction* and *Fourier Graph Neural Network (FGNN)*.

3.1 Problem Formulation

Following previous works [11, 13, 29], we define each social event as a quadruple (s, r, o, t) , where $s \in \mathcal{E}$, $r \in \mathcal{R}$ and $o \in \mathcal{E}$ represent subject entity, relation and object entity, respectively. t denotes the timestamp when the event occurs, and $\mathcal{E}$ and $\mathcal{R}$ are the universal sets of entities and relations. Given a historical window size τ , all quadruples occurring between $(t - \tau, t)$ are viewed as historical events, and our aim is to predict the quadruples in timestamp $t + 1$. We focus on two subtasks: entity prediction subtask that predicts the missing object o of query $(s, r, [\text{mask}], t + 1)$ or missing subject s of query $([\text{mask}], r, o, t + 1)$, and link prediction subtask, which aims to determine the missing relation r based on query $(s, [\text{mask}], o, t + 1)$.

3.2 Frequency-domain Theorems

Fast Fourier Transformation (FFT) is an efficient implementation used to represent signals in the frequency domain, so as to analyze the different frequency components. In time series analysis, the Real FFT (rFFT) is commonly used for processing real-valued input signals. It transforms an input of n real numbers into a sequence of $N/2 + 1$ complex numbers (N represents the sequence length), effectively representing the signal in the complex frequency domain.

The Convolution Theorem reveals the relationship between convolution operations and the Fourier transform. To be specific, if two functions $f(t)$ and $g(t)$ have Fourier transforms $F(\omega)$ and $G(\omega)$ respectively, then their convolution in the time domain corresponds to the product in the frequency domain, and vice versa.

4 Method

In this section, we presents the detailed workflow of FGL as shown in Figure 2. The pseudo code can be checked in **Appendix B**.

4.1 Fourier Graph (FG)

The Fourier Graph construction stage aims to reconstruct the input event graphs sequence $G_{(t-\tau, t)}$ ranging from timestamp $t - \tau$ to

t . It outputs corresponding FG G_t . This stage contains a complex-valued event encoder and a Fourier graph constructor.

4.1.1 Complex-valued event encoder. The encoder contains a real encoder for real part of complex vectors, an imaginary encoder producing distinct (low, middle, high) frequency component representations, and a semantic-aware frequency filter extracting semantic-specific temporal patterns. Firstly, we employ a real encoder to embed nodes s, o and edges r in graph sequence $G_{(t-\tau, t)}$ into semantic representations e_s, e_o and e_r , as shown in Figure 2(a):

$$e_s = E(s), e_o = E(o), e_r = E(r) \quad s, o, r \in G_{(t-\tau, t)} \quad (1)$$

where $E(\cdot)$ is an embedding layer, which embeds the nodes and entity into high dimension semantic representations $e_s, e_o, e_r \in \mathbb{R}^{d_e}$. Secondly, we employ a imaginary encoder to generate raw frequency representations for each timestamp, as shown in Figure 2(b). We apply a non-trainable periodic encoding for timestamp $t' \in [t - \tau, t]$, where the obtained embedding $P^{(t')}$ is expressed as:

$$\begin{aligned} P_{2p}^{(t')} &= \sin\left(\frac{t'}{10000^{2p/d_t}}\right) \\ P_{2p+1}^{(t')} &= \cos\left(\frac{t'}{10000^{2p/d_t}}\right) \\ P^{(t')} &= [P_0^{(t')}, P_1^{(t')}, \dots, P_{d_t-1}^{(t')}] \end{aligned} \quad (2)$$

where d_t denotes the embedding size of periodic encoding. $P^{(t')} \in \mathbb{R}^{d_t}$ represents the periodic codes generated for timestamp t' . $P_{2p}^{(t')}$ and $P_{2p+1}^{(t')}$ represent embeddings for the even and odd position. Then, we utilize rFFT to transform $P^{(t')}$ into frequency domain:

$$\begin{aligned} \mathbf{P}_k^{(t')} &= \sum_{i=0}^{d_t-1} P_i^{(t')} \cdot e^{-j \frac{2\pi k i}{d_t}} \quad k \in \{0, 1, \dots, \lfloor d_t/2 \rfloor\} \\ \mathbf{P}^{(t')} &= [\mathbf{P}_0^{(t')}, \mathbf{P}_1^{(t')}, \dots, \mathbf{P}_{\lfloor d_t/2 \rfloor}^{(t')}] \end{aligned} \quad (3)$$

where j is the imaginary unit, $\mathbf{P}_k^{(t')} \in \mathbb{C}^{\lfloor d_t/2 \rfloor}$ is the produced frequency domain representation. However, this representation is

usually very dense and noisy. To filter noises, we first split the frequency representation into low, middle, high frequency components according to the cutoff frequencies k_{c1} and k_{c2} :

$$\begin{aligned} \mathbf{P}_{LF}^{(t')} &= \{\mathbf{P}_k^{(t')} \mid 0 \leq k < k_{c1}\} \\ \mathbf{P}_{MF}^{(t')} &= \{\mathbf{P}_k^{(t')} \mid k_{c1} \leq k < k_{c2}\} \\ \mathbf{P}_{HF}^{(t')} &= \{\mathbf{P}_k^{(t')} \mid k_{c2} \leq k \leq \lfloor d_t/2 \rfloor\} \end{aligned} \quad (4)$$

where the low frequency component feature $\mathbf{P}_{LF}^{(t')}$ corresponds to slower, longer-term trends with extended periods specified to timestamp t' , while higher frequency component feature $\mathbf{P}_{HF}^{(t')}$ corresponds to faster, shorter-term dynamics with shorter periods on t' . Furthermore, to keep the frequency representations concise, we perform shrinkage and normalization on $\mathbf{P}_{LF}^{(t')}$, $\mathbf{P}_{MF}^{(t')}$, $\mathbf{P}_{HF}^{(t')} \in \mathbb{C}^{\lfloor d_t/6 \rfloor}$ to filter out minor frequency components.

Thirdly, semantic-aware frequency filter adaptively select frequency components $\mathbf{P}_{LF}^{(t')}$, $\mathbf{P}_{MF}^{(t')}$, $\mathbf{P}_{HF}^{(t')}$, as shown in Figure 2(c). This filtering process is parameterized by semantics e_s, e_r, e_o :

$$W_*^f = f_W(e_*), B_*^f = f_B(e_*) \quad * \in \{s, o, r\} \quad (5)$$

$$\mathbf{P}^{(t',*)} = \sigma(W_*^f \cdot (\mathbf{P}_{LF}^{(t')} \oplus \mathbf{P}_{MF}^{(t')} \oplus \mathbf{P}_{HF}^{(t')} + B_*^f)) \quad (6)$$

where σ refers to activation function, $\oplus$ denotes stack operation, and $\langle \cdot \rangle$ is the average operation along the new dimension after stacking. W_*^f and B_*^f are parameters derived from parameter generation networks $f_W(\cdot)$ and $f_B(\cdot)$. $\mathbf{P}^{(t',*)} \in \mathbb{C}^{\lfloor d_t/6 \rfloor}$ are filtered temporal patterns specified to subject entity semantics, object entity semantics and relations semantics, respectively.

After obtaining the filtered temporal patterns $\mathbf{P}^{(t',*)}$, we transform them back into time domain via irFFT:

$$\begin{aligned} \mathbf{P}^{(t',*)} &= [\mathbf{P}_0^{(t',*)}, \mathbf{P}_1^{(t',*)}, \dots, \mathbf{P}_{\lfloor d_t/6 \rfloor}^{(t',*)}] \\ P_n^{(t',*)} &= \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{P}_k^{(t',*)} \cdot e^{i \frac{2\pi k n}{N}} \\ P^{(t',*)} &= [P_0^{(t',*)}, P_1^{(t',*)}, \dots, P_{N-1}^{(t',*)}] \end{aligned} \quad (7)$$

where $N = \lfloor d_t/3 \rfloor$ is the number of DFT points, $P^{(t',*)} \in \mathbb{R}^{\lfloor d_t/3 \rfloor}$ is the reconstructed temporal patterns in the time domain specified to semantic $*$ at timestamp t' . Moreover, we propose a date encoder, which add date-related periodic information to $P^{(t',*)}$:

$$\begin{aligned} e_t &= \left[\sin\left(\frac{2\pi t}{T_j}\right), \cos\left(\frac{2\pi t}{T_j}\right) \right] \quad T_j \in \{365, 30, 7\} \\ z^{(t',*)} &= P^{(t',*)} \parallel e_{t'} \quad * \in \{s, r, o\} \end{aligned} \quad (8)$$

where e_t is the date embedding for timestamp t , $\parallel$ refers to concatenation operation, $T_j \in \{365, 30, 7\}$ is the periods for yearly encoding, monthly encoding and weekly encoding. $z^{(t',*)}$ is the produced final temporal patterns specified by semantic $*$ and timestamp t' . The date encoder is proposed to provide additional temporal contexts and positional information for temporal patterns $P^{(t',*)}$. More explanations on temporal patterns are presented in **Appendix G.1**.

4.1.2 Fourier Graph Constructor. Based on semantics and temporal patterns, we construct FG as shown in the Figure 2(d).

Definition 1. Fourier Graph. Given a time window $[t - \tau, t]$, a *Fourier Graph* can be defined as $\mathcal{G}_t = \{\mathcal{E}, \mathcal{R}\}$. $\mathcal{E}$ refers to the entity set, $\mathcal{R}$ refers to the relation set. The representation of subject $s \in \mathcal{E}$, object $o \in \mathcal{E}$, relation $r \in \mathcal{R}$ can be expressed as:

$$\mathbf{s} = e_s + jz^{(t,s)} \quad \mathbf{o} = e_o + jz^{(t,o)} \quad \mathbf{r} = e_r + jz^{(t',r)} \quad (9)$$

where $\mathbf{s}, \mathbf{o}, \mathbf{r} \in \mathbb{C}^{de}$, $t - \tau \leq t' \leq t$ is a specific timestamp in the time window, $\mathbb{C}$ is the algebra in complex plane. Note here we assign different timestamps to entities and relations in one event. This comes with two reasons: (1) entities tend to be consistent for a long time, while the occurrence of relation has a specific time point. To this end, for time window $[t - \tau, t]$, we directly assign entities with the maximum timestamp t and relations with their occurrence time t' . (2) We aim to introduce time difference information. Assigning t to node and t' to edge explicitly encodes a time difference $\Delta t = t - t'$, which directly reflects recency of events. More discussions on Fourier Graph are provided in **Appendix G.2**.

4.2 Fourier Graph Neural Network (FGNN)

4.2.1 Fourier Space Multiplier. As both entities $\mathbf{s}, \mathbf{o}$ and relations $\mathbf{r}$ are complex vectors, according to the rule of multiplication of complex numbers, we use frequency domain multiplication as aggregation in FG, as shown in Figure 2(e). The multiplication for node prediction subtask is:

$$\begin{aligned} \mathbf{s} \times \mathbf{r} &= C_1 + jC_2 + jC_3 - C_4 \\ &= e_s e_r + j e_s z^{(t',r)} + j z^{(t,s)} e_r - z^{(t,s)} z^{(t',r)} \end{aligned} \quad (10)$$

where C_1 leads to semantic fusion, C_4 is a penalty term used to indicate the matching degree of temporal patterns between $\mathbf{s}$ and $\mathbf{r}$. jC_2 and jC_3 involve interactions between semantics and temporal patterns. For node prediction subtask, we multiply $\mathbf{s}$ and $\mathbf{o}$ instead.

The core of Fourier space multiplier is utilizing frequency-domain multiplication to substitute time-domain spatio-temporal convolutions. We theoretically analyze this replacement as follows:

Theorem 1. Generalized Convolution Theorem. Given time-domain temporal convolution kernel function $f(t)$ and spatial convolution kernel function $g(t)$, their combination $f(t) * g(t)$ is equivalent to frequency-domain multiplication $F(\omega) \cdot G(\omega)$ as follows:

$$\mathcal{F}\{g * (f * x)\} = G(\omega) \cdot F(\omega) \cdot X(\omega) \quad (11)$$

where $\mathcal{F}$ refers to Fourier transform, $*$ refers to convolutions, x and $X(\omega)$ are the inputs in time domain and frequency domain.

Also, we provide a theoretical proof demonstrating the equivalence of total energy in the time and frequency domain, which means the information loss during time-domain and frequency-domain transformations is negligible:

Theorem 2. Discrete Parseval's Theorem. For a discrete signal $x(n)$ and its discrete Fourier transformation $X[k]$, the total energy of the signal is equal whether in time or frequency domain:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (12)$$

The proofs for Theorem 1&2 are provided in **Appendix C.1, C.2**.

4.2.2 Complex-valued Event Decoder. After the multiplication, we can obtain four components as shown in Eq.(10). As shown in Figure 2(f), to better extract key information from these components, we first introduce two specific loss functions to further adjust the learned feature space. For components containing semantics like $e_s e_r$, $j e_s z^{(r)}$ and $j z^{(s)} e_r$, we utilize the 3-norm operation to ensure that learned semantics are neither too sparse nor overly smooth, thereby preventing the collapse of the feature space:

$$\mathcal{L}_r = \sum (\|C_1\|_3^3 + \|C_2\|_3^3 + \|C_3\|_3^3) \quad (13)$$

where $\|\cdot\|_3$ stands for the 3-norm operation.

Moreover, to improve the learning of temporal patterns, we further utilize the temporal difference regularization loss, which makes the embedding slowly change over time. The regularized embeddings from adjacent timestamps should be close. This loss is applied to components $j e_s z^{(r)}$, $j z^{(s)} e_r$ and $z^{(s)} z^{(r)}$ as follows:

$$\mathcal{L}_t = \frac{1}{t} \sum_{i=1}^t \left\| C_2^{(t'+1)} - C_2^{(t')} \right\|_4^4 + \left\| C_3^{(t+1)} - C_3^{(t)} \right\|_4^4 + \left\| C_4^{(t'+1, t+1)} - C_4^{(t', t)} \right\|_4^4 \quad (14)$$

For simplicity, we append superscript (t') and (t) for C_2, C_3, C_4 to mark the timestep of $z^{(t', r)}$, $z^{(t, s)}$.

Furthermore, we develop a novel **Real-Imaginary Similarity (RIS)** score as scoring function to rank candidates, which is shown in Figure 2(g). Taking the entity prediction subtask as an example, we first group four components with adaptive weights and form the target object complex vector X_o , and further measure the similarity between X_o and candidates X_c :

$$X_o = e_o + j z_o = W_1 C_1 - W_4 C_4 + j W_2 C_2 + j W_3 C_3 \quad (15)$$

$$\text{RIS}(X_o, X_c) = W_5 \text{R}(X_o) \text{R}(X_c) + W_6 \text{I}(X_o) \text{I}(X_c) - \omega (\|X_o\|^2 + \|X_c\|^2) \quad (16)$$

where $\text{R}(\cdot)$ and $\text{I}(\cdot)$ are real and imaginary part of complex vectors, ω is the weight for penalty term. The aim of this scoring function is to measure the similarity between target complex features and candidate complex features. A higher score indicates that the quadruple is more compatible with the candidate, and also suggests that the corresponding event is more likely to occur.

4.3 Optimization Target

The loss function contains categorical cross-entropy loss for evaluating the positive or negative facts, a 3-norm reg loss $\mathcal{L}_r$ and a temporal diff reg loss $\mathcal{L}_t$. Their combinations are shown as follows:

$$\mathcal{L} = \sum_Q \left[-\log \left(\frac{\exp(Q^p(s, o, r, t))}{\sum \exp(Q^n(s, o, r, t))} \right) \right] + \lambda_1 \mathcal{L}_r + \lambda_2 \mathcal{L}_t \quad (17)$$

where Q refers to all quadruples in the dataset, Q^p and Q^n are positive and negative samples of quadruples, λ_1 and λ_2 are adaptive weights of different loss functions.

Table 1: Detailed statistics of used GDELT dataset.

Dataset	$ \mathcal{V} $	$ \mathcal{E} $	#Days	#Train	#Valid	#Test
EG	2,594	225	2,584	377,430	36,588	28,644
IR	2,988	236	2,584	973,752	69,827	76,239
IS	3,456	238	2,584	1,430,389	171,518	156,695
ME	8,622	342	2,584	25,133,953	3,141,744	2,929,510
EUR	11,656	362	2,584	38,223,620	4,778,019	4,567,119

5 Experiments

In this section, we aim to answer the following research questions:

- **RQ1:** How does FGL perform on mainstream benchmarks?
- **RQ2:** What is the individual contribution of each component to FGL’s overall performance?
- **RQ3:** Can FGL satisfy the real-time and computational complexity requirements inherent to web-based event streams?
- **RQ4:** How does FGL’s performance compare to that of large language model (LLM)?
- **RQ5:** How does FGL enhance interpretability?

5.1 Dataset Details

The experiments are conducted on three real-world event datasets collected from Global Database of Events, Language, and Tone (GDELT)¹. The GDELT Event Database [23] is a global web news ecosystem, recording over 300 categories of physical activities throughout the World Wide Web. Following the mainstream benchmarks [13, 14], we crop three subsets of GDELT according to the regions of the events: Egypt (EG), Iran (IR), and Israel (IS). The time period spans from February 2015 to March 2022, which is further divided into training, validation, and testing sets in an 8:1:1 ratio over the timeline. Specifically, the training set covers the period from February 2015 to February 2021, the validation set spans from March 2021 to August 2021, and the test set ranges from August 2021 to March 2022. This test set is later than the knowledge cutoff of the GPT-3.5 Turbo model², thus avoiding potential data leakage issues with the LLM model. To further demonstrate FGL’s efficiency on real world applications, we additionally collect two very large regional datasets: ME (Middle East) and EUR (Europe). These regional datasets encompass multiple countries and significantly more quadruples than mainstream benchmarks, which pose greater challenges. The details are demonstrated in Table 1.

5.2 Evaluation Protocols

We evaluate our model on both entity prediction and relation prediction subtasks. In the dataloader, we replace elements to be predicted with all elements from $\mathcal{E}$ or $\mathcal{R}$ in turn for each quadruple in the test set. We then use time-wise filtered setting to score all corrupted quadruples and rank candidate elements based on scores. Positive candidates should have a higher ranking priority than negative ones. Following the metrics in previous studies [11, 13, 14], we use mean reciprocal rank (MRR) and HIT@1,3,10 as our evaluation

¹<https://www.gdeltproject.org/>

²<https://platform.openai.com/docs/models/gpt-3.5-turbo>

Table 2: The results for entity prediction subtask. The best and sub-optimal models are marked by boldface and underline.

Model		EG				IR				IS			
		MRR	HIT@1	HIT@3	HIT@10	MRR	HIT@1	HIT@3	HIT@10	MRR	HIT@1	HIT@3	HIT@10
DistMult	<i>ICLR'15</i>	11.64	3.44	12.14	29.27	13.49	3.92	14.68	33.79	10.31	2.33	9.29	29.50
ConvE	<i>AAAI'18</i>	11.51	3.12	12.72	28.82	13.65	4.09	14.85	34.00	10.60	2.51	9.84	29.35
ConvTE	<i>AAAI'19</i>	12.41	4.58	17.32	29.21	14.05	4.62	15.29	34.12	10.79	2.87	9.94	29.30
RE-NET	<i>EMNLP'20</i>	12.12	4.13	12.24	29.32	14.01	4.51	15.01	34.52	10.64	2.63	10.16	28.94
TANGO	<i>EMNLP'21</i>	10.43	2.40	11.06	27.61	12.49	2.81	13.67	33.14	9.72	1.71	8.52	28.89
RE-GCN	<i>SIGIR'21</i>	12.45	3.52	13.66	31.01	16.47	6.22	17.96	38.38	13.01	4.08	12.81	33.46
EvoKG	<i>WSDM'22</i>	7.97	1.20	7.75	25.29	8.92	0.11	7.67	31.20	7.79	0.08	5.18	27.89
GTRL	<i>TKDE'24</i>	13.38	3.75	15.68	33.00	18.67	6.93	15.99	38.88	15.78	4.04	12.48	31.17
RotatE	<i>ICLR'19</i>	8.92	1.25	7.72	27.48	10.55	1.25	10.74	31.52	8.79	1.32	6.52	23.07
ConvRot	<i>ESWA'24</i>	8.63	1.11	6.95	25.32	9.41	1.62	10.83	31.59	8.51	1.11	6.20	24.65
DCNE	<i>COLING'24</i>	9.02	1.75	8.52	26.84	10.17	2.69	10.48	30.36	10.73	2.01	7.52	26.64
SeCoGD	<i>KDD'23</i>	14.64	<u>5.93</u>	<u>16.05</u>	32.36	17.57	7.24	<u>19.02</u>	39.75	15.52	5.95	15.88	36.93
DHyper	<i>TOIS'23</i>	<u>16.57</u>	5.25	15.67	31.89	<u>18.66</u>	<u>8.07</u>	<u>18.57</u>	<u>40.27</u>	<u>16.68</u>	<u>6.01</u>	<u>16.71</u>	<u>38.50</u>
FGL	<i>This paper</i>	24.51	14.87	25.96	41.31	26.64	18.73	27.56	50.31	28.51	18.67	27.38	48.00
Δ_{sota} (%)		7.94	8.94	9.91	8.31	7.98	10.66	8.54	10.04	11.83	12.66	10.67	8.50

metrics. MRR is employed as the main metric. We provide more implementation details in the **Appendix D**.

5.3 Baselines

We consider five kinds of models as baselines: (1) Static Graph Based Methods treat social event prediction as link prediction task on a static knowledge graph, such as: ConvE [5], ConvTransE [24], DistMult [32]. (2) Temporal Graph Based Methods view historical timelines as dynamic graphs. We consider the following baselines: TANGO [7], RE-GCN [11], RE-NET [8], EvoKG [18], GTRL [28] and HisMatch [12]. (3) Fourier or Spectral Based Methods employ Fourier or spectral techniques to enhance the learning of event embeddings, including RotatE [27], ConvRot [9], DCNE [6]. (4) Hypergraph Based Methods capture long-term temporal dependencies via hyperedges to directly connect entities in different timestamps, such as SeCoGD [13] and DHyper [29]. (5) LLM Based Methods utilize large language models as forecasters [26, 35]. We consider ChatGPT-3.5-Turbo as the representative baseline.

5.4 Mainstream Benchmarks (RQ1)

Table 2 shows the mainstream benchmark results for the entity prediction subtask. We highlight three key findings here: (1) FGL outperforms all baselines on all three datasets. Specifically, FGL significantly outperforms static graph and temporal graph based methods. While hypergraph based methods achieve sub-optimal performance, FGL still manages to stable performance advantages. This verifies FGL’s effectiveness. (2) FGL performs better on larger datasets. Among three mainstream datasets in GDEL, EG contains the least number of nodes and relations, while IS has the most. Correspondingly, FGL achieves 8.50%-12.66% performance gain, which is larger than those in EG dataset. We argue that more entities and relations allow FGL to construct a much denser FG structure. With more entities and relations, there are more opportunities for recurring patterns to emerge and be captured by Fourier graph

Table 3: The results for relation prediction subtask. Boldface and underline represents the best and sub-optimal MRR.

Model		EG	IR	IS
ConvE	<i>AAAI'18</i>	17.81±0.13	18.77±0.15	20.60±0.21
ConvTE	<i>AAAI'19</i>	18.97±0.11	18.81±0.09	20.73±0.14
RE-GCN	<i>SIGIR'21</i>	<u>21.44±0.05</u>	<u>23.94±0.21</u>	<u>22.84±0.31</u>
RE-NET	<i>EMNLP'20</i>	20.42±0.13	19.67±0.17	22.14±0.18
HisMatch	<i>EMNLP'22</i>	18.51±0.09	20.36±0.15	21.84±0.04
FGL	<i>This paper</i>	29.95±0.09	30.61±0.15	30.93±0.03
Δ_{sota} (%)		7.37	5.67	8.23

constructor, leading to better performance. (3) Traditional Fourier or spectral based methods perform poorly in the benchmark. Note that these methods utilize Fourier-related techniques to deal with asymmetric relations and relation rotations in static knowledge graphs. However, they fail to employ Fourier transformation for intricate temporal dependencies. This significantly harms their performance of social event prediction in real-world environments.

For the relation prediction subtask, results are presented in Table 3. As some baseline models are not applicable to this subtask, we present MRR results for a selection of representative baselines. Note that the number of relation types is far smaller than the number of node types in GDEL dataset. This makes the relation prediction simpler, leading to higher average performance across all models compared to the entity prediction. The performance of FGL demonstrates that our proposed Fourier space multiplier enables flexible switch between node prediction and link prediction tasks. Many existing models (such as RGCN, TransE, and DistMult) use different representation spaces or objective functions to separately optimize nodes and edges. In contrast, our method represents and optimizes nodes and edges in a unified manner, leading to better performance.

Table 4: Results of ablation studies.

Model	EG		IR		IS	
	MRR	HIT@1	MRR	HIT@1	MRR	HIT@1
RE-GCN	12.45	3.52	16.47	6.22	13.01	4.08
SeCoGD	14.64	5.93	17.57	7.24	15.52	5.95
FGL	24.51	14.87	26.64	18.73	28.51	18.67
w/o temporal	16.16	7.23	18.17	10.83	20.25	9.15
w/o date	22.21	12.57	23.67	13.55	25.71	16.66
w/o freq splitter	23.41	12.65	25.29	11.91	23.86	13.12
w/o $\mathcal{L}_t$	20.15	13.10	24.15	15.93	27.18	15.15
w/o $\mathcal{L}_r$	19.67	12.07	18.64	10.34	13.63	7.53
w/o RIS	19.87	12.03	22.67	13.45	24.57	21.53

Table 5: Results of wall-clock time comparison, which includes training time per epoch and total inference time.

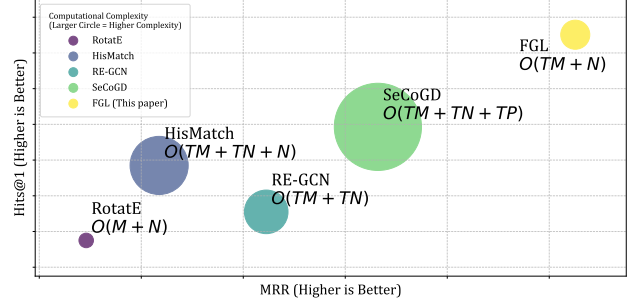
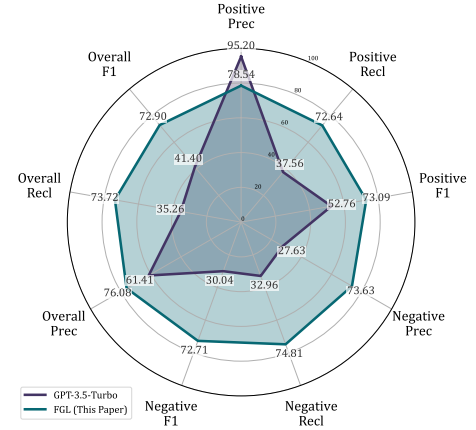
Model	EG Train (s/epoch)	EG Infer (s)	IR Train (s/epoch)	IR Infer (s)	IS Train (s/epoch)	IS Infer (s)
HisMatch	289.1	24.1	433.1	35.8	589.7	53.6
RE-GCN	230.2	20.8	398.1	29.6	454.7	49.9
SeCoGD	313.6	21.4	568.2	35.3	629.8	54.1
FGL	40.9	25.1	89.7	33.3	129.8	40.9

5.5 Ablation Study (RQ2)

To further show the contribution of each individual component, we create several variants as shown in Table 4: (1) w/o temporal removes temporal patterns $P^{(t',*)}$ from the imaginary part of nodes and edges representations, leaving only date embedding e_t from the date encoder ($z^{(t',*)} = e_t$); (2) w/o date removes date embedding e_t from the imaginary part of nodes and edges, leaving only the temporal patterns ($z^{(t',*)} = P^{(t',*)}$); (3) w/o freq splitter directly utilizes full frequency representation without splitting into low, mid and high components; (4) w/o $\mathcal{L}_r$ removes the 3-norm loss; (5) w/o $\mathcal{L}_t$ removes the temporal difference regularization loss; (6) w/o RIS removes RIS score and utilize a simple MLP to bundle complex representations for ranking scores. We highlight some findings here: Removing temporal patterns (w/o temporal) from imaginary part leads to a significant performance drop, demonstrating the effectiveness of incorporating periodic information. Removing date embeddings (w/o date) leads the loss of positional information and harms model’s performance. Similarly, removing the frequency splitter leads to noisy frequency-domain embeddings and causes unstable performance. Removing 3-norm loss (w/o $\mathcal{L}_r$) and temporal difference regularization loss (w/o $\mathcal{L}_t$) lead to substantial performance drop. We argue this phenomenon stems from the collapse of the feature space. Without these regularization, the features inevitably cluster around a small subset of components in the space, leading to excessive similarity among feature vectors.

5.6 Complexity Study (RQ3)

To demonstrate the computational efficiency in web-based scenario, we design complexity and wall-clock time comparison with four representative baselines: HisMatch [12], RE-GCN [11], RotatE [27]

**Figure 3: Theoretical time complexity comparison with representative baselines. T refers to historical window size, M and N refer to daily edge number and node number.****Figure 4: The benchmark results with GPT-3.5-Turbo. We align the testing set (Aug 2021) with knowledge cutoff date of GPT-3.5-Turbo (Sep 2021) to avoid data leakage.**

and SeCoGD [13], which represent temporal graph based methods, Fourier based methods and hypergraph based methods.

We visualize the theoretical computational costs in Figure 3 (Calculation is provided in **Appendix E**). Moreover, we provide a wall-clock time comparison in Table 5. We highlight two findings here: (1) FGL significantly reduces the time required for model training, whose speedups reach 5.6-7.7 $\times$ on EG, 4.8-6.3 $\times$ on IR and 3.5-4.8 $\times$ on IS, confirming FGL’s high training efficiency. (2) In terms of inference speed, FGL varies with datasets. We argue that initializing and running rFFT may slightly increase inference delay.

5.7 Comparison with LLM (RQ4)

To comprehensively evaluate our proposed model, we further compare FGL with the large language models. It is worth noting that the evaluation protocol of existing LLM based methods is different from that of graph based methods. In LLM based methods, the task is usually defined as a binary classification issue, where the objective is to predict the possibility of a specific event happen in the future [35]. To align with these methods, we further tweak the output function of FGNN from ranking issue to binary classification. Moreover, to

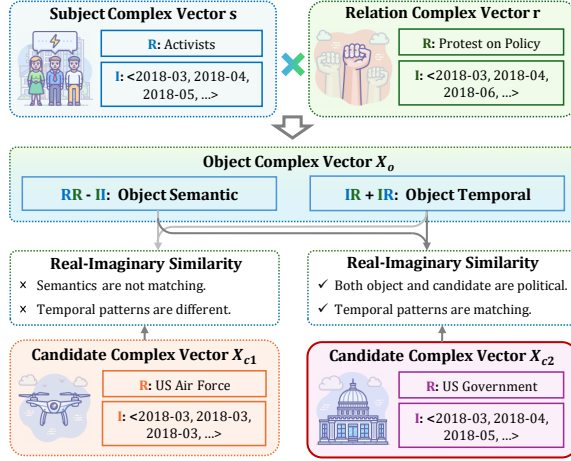


Figure 5: Case study of FGL, which shows a concrete example for quadruple (Activists, Protest on Policy, US Government, 2022-02-17) from test set of IS dataset.

avoid the data leakage problem, we utilize the ChatGPT-3.5-Turbo model as the backbone of LLM baselines. The knowledge cutoff of ChatGPT-3.5-Turbo model is earlier than the division of validation and testing set, ensuring that the evaluated baselines have not seen any information from test sets during training.

The Results in Figure 4 show that LLMs struggle with the complex temporal and relational patterns of social events. They perform poorly on negative samples (fabricated events), frequently incorrectly classifying them as positive. We attribute this to the hallucination problem inherent in LLMs [3, 26], where they generate false information, which severely undermines their reliability and credibility for decision-making. Moreover, the potential data leakage issue remains a non-trivial challenge for LLMs to perform a fair comparison, which remains a promising future work direction. In contrast, FGL achieves a more balanced and superior performance across both positive and negative samples, which makes FGL more reliable for decision making.

5.8 Case Study and Interpretability Study (RQ5)

In this subsection, we demonstrate a concrete example to show the interpretability of FGL as shown in Figure 5. Specifically, we choose the quadruple (Activists, Protest on Policy, US Government, 2020-05-01) from the testing set of IS dataset. In our model, a subject’s complex vector is multiplied by a relation’s complex vector to form a target object vector. This vector uniquely encodes semantics in its real part and temporal patterns in its imaginary part. A Real-Imaginary Similarity (RIS) score then measures the match between this predicted object vector and potential candidates.

To further analyze the model’s interpretability, we visualize the learned temporal patterns in the frequency domain, as shown in Figure 6. This visualization highlights two key advantages of our approach: (1) The learning process of temporal patterns is interpretable. For instance, the frequency spectrum for the ‘Protest on Policy’ relation is concentrated in low-to-mid frequencies, correctly identifying its tendency for long-term recurrence. More strikingly,

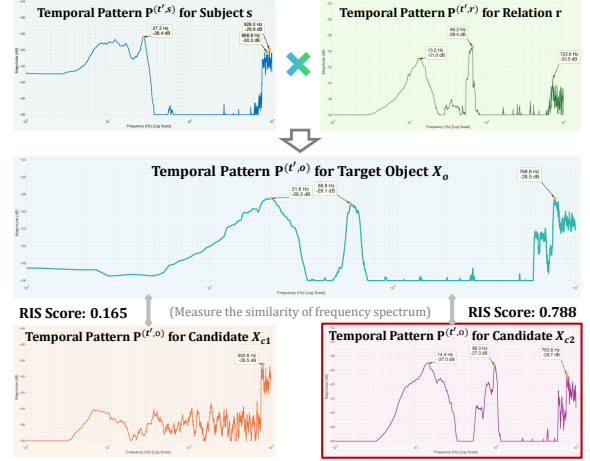


Figure 6: The frequency spectrum of temporal patterns of entities and relations presented in Figure 5. The x-axis is log-transformed frequency and y-axis is amplitude.

the ‘Activist’ subject, known for monthly recurrence, displays a prominent spectral peak at a frequency corresponding to a 30-day period. This alignment confirms that our model accurately captures the underlying temporal dynamics of events. (2) Frequency-domain theory serves as interpretable backbone for utilizing temporal pattern. Current methods obtain refined features via frequency techniques and map them (MLP, Transformer) to final output [2, 33, 34], which obscures the original spectral information. In contrast, FGL preserves these components using complex-valued embeddings. It then employs the convolution theorem (FGNN) to combine the subject and relation spectra into the target object’s spectrum via an interpretable manner. Finally, our proposed RIS scoring function directly compares these frequency spectra to find the best match, making the prediction process transparent and interpretable.

6 Conclusion

This paper introduces Fourier Graph Learning (FGL), a novel frequency domain paradigm for social event prediction that addresses the limitations of capturing long-term temporal dependencies and managing computational complexity. It contains Fourier Graph (FG), a novel data structure where nodes and edges are complex vectors, integrating semantics and semantic-specific temporal patterns. Building upon FG, the Fourier Graph Neural Network (FGNN) leverages efficient frequency-domain multiplication for aggregation and a complex-valued event decoder for prediction. Extensive experiments on real-world GDELT datasets demonstrate FGL’s superior performance in accuracy, computational efficiency, and interpretability compared to baselines. We will proceed to further explore the prediction of rare events in the future.

Acknowledgments

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A Notations

All used notations in the paper are presented in Table 6.

Table 6: Notations and Descriptions

Notations	Definitions and Descriptions
$G_{(t-\tau, t)}$	Historical graph sequences from $t - \tau$ to t
$\mathcal{G}_t$	Fourier Graph (FG) at t
e_s, e_o, e_r	Semantics of subject, object and relations
$P^{(t')}$	Time domain periodic embedding for t'
$\mathbf{P}^{(t')}$	Frequency domain periodic embedding for t'
$\mathbf{P}_{LF}^{(t')}, \mathbf{P}_{MF}^{(t')}, \mathbf{P}_{HF}^{(t')}$	The low, middle and high frequency components derived from $\mathbf{P}^{(t')}$
$* \in \{s, o, r\}$	Refers to any of subject, object and relation
$\mathbf{P}^{(t',*)}$	Semantic-specific temporal patterns in frequency domain
$P^{(t',*)}$	Semantic-specific temporal patterns in time domain
$z^{(t',*)}$	Temporal patterns with date embeddings
$\mathbf{s}, \mathbf{o}, \mathbf{r}$	Complex-valued subject, object, relations in FG $\mathcal{G}_t$
$\mathbf{X}_o$	Target object complex vector
$\mathbf{X}_c$	Candidate object complex vector

B Pseudo Code

The pseudo code for Section 4.1 Section 4.2 are presented in Algorithm 2 and Algorithm 1, respectively.

Algorithm 1 Pseudo Code for Fourier Graph Neural Network

Require: Fourier Graph $\mathcal{G}_t$
Ensure: Target object $\mathbf{o}$ for query $(\mathbf{s}, \mathbf{r}, [\text{mask}])$ in $\mathcal{G}_t$

```

1: function FOURIERGNN
2:   for all queries  $(\mathbf{s}, \mathbf{r}, [\text{mask}]) \in \mathcal{G}_t$  do
     // Fourier space multiplier
3:      $C_1 = \mathbf{R}(\mathbf{s}) \mathbf{R}(\mathbf{r}); C_2 = \mathbf{R}(\mathbf{s}) \mathbf{I}(\mathbf{r})$ 
4:      $C_3 = \mathbf{I}(\mathbf{s}) \mathbf{R}(\mathbf{r}); C_4 = \mathbf{I}(\mathbf{s}) \mathbf{I}(\mathbf{r})$  ▷ Eq.(10)
     // Complex-valued Event Decoder
5:      $\mathcal{L}_r \leftarrow 3\text{-norm}(C_1, C_2, C_3)$  ▷ Eq.(13)
6:      $\mathcal{L}_t \leftarrow \text{TimeDiff}(C_2, C_3, C_4)$  ▷ Eq.(14)
7:      $\mathbf{X}_o \leftarrow \text{Group}(C_1, C_2, C_3, C_4)$  ▷ Eq.(15)
8:     for all candidates  $\mathbf{X}_c$  do
9:        $S \leftarrow \text{RIS}(\mathbf{X}_c, \mathbf{X}_o)$  ▷ Eq.(16)
10:      If  $S$  is Max then  $\mathbf{o} \leftarrow \mathbf{X}_c$ 
11:    end for
12:  end for
13:  return  $\mathbf{o}, \mathcal{L}_t$  and  $\mathcal{L}_r$ 
14: end function
```

Algorithm 2 Pseudo Code for Fourier Graph construction

Require: Event graph sequence $G_{(t-\tau, t)}$
Ensure: Fourier Graph $\mathcal{G}_t$

```

1: function FOURIERGRAPH
   // 1) Complex-valued event encoder
   // Real encoder
2:   for all entities  $s, o$  and edges  $r$  in  $G_{(t-\tau, t)}$  do
3:      $e_s \leftarrow E(s); e_o \leftarrow E(o); e_r \leftarrow E(r)$  ▷ Eq.(1)
4:   end for
   // Imaginary encoder
5:   for all timestamp  $t' \in [t - \tau, t]$  do
6:      $P^{(t')} \leftarrow \text{PeriodicEncode}(t', d_t)$  ▷ Eq.(2)
7:      $\mathbf{P}^{(t')} \leftarrow \text{rFFT}(P^{(t')})$  ▷ Eq.(3)
8:      $\mathbf{P}_{LF}^{(t')}, \mathbf{P}_{MF}^{(t')}, \mathbf{P}_{HF}^{(t')} \leftarrow \text{Split}(\mathbf{P}^{(t')})$  ▷ Eq.(4)
     // Semantic-aware frequency filter
9:     for all semantic  $* \in \{s, o, r\}$  do
10:       $F \leftarrow \text{InitFilter}(e_*)$  ▷ Eq.(5)
11:       $\mathbf{P}^{(t',*)} \leftarrow F(\mathbf{P}_{LF}^{(t')}, \mathbf{P}_{MF}^{(t')}, \mathbf{P}_{HF}^{(t')})$  ▷ Eq.(6)
12:       $P^{(t',*)} \leftarrow \text{irFFT}(\mathbf{P}^{(t',*)})$  ▷ Eq.(7)
13:    end for
14:     $z^{(t',*)} \leftarrow \text{DateEmb}(P^{(t',*)})$  ▷ Eq.(8)
15:  end for
   // 2) Fourier Graph constructor
16:   $\mathcal{E} \leftarrow \emptyset; \mathcal{R} \leftarrow \emptyset$ 
17:  for all entities  $s, o$  and edges  $r$  in  $G_{(t-\tau, t)}$  do
18:     $\mathbf{s} = e_s + jz^{(t,s)}; \mathbf{o} = e_o + jz^{(t,o)}$ 
19:     $\mathbf{r} = e_r + jz^{(t,r)}$  ▷ Eq.(9)
20:    Add  $\mathbf{s}, \mathbf{o}$  to  $\mathcal{E}$ ; Add  $\mathbf{r}$  to  $\mathcal{R}$ 
21:  end for
22:  return  $\mathcal{G}_t = \{\mathcal{E}, \mathcal{R}\}$ 
23: end function
```

C Proofs

C.1 Proof of Generalized Convolution Theorem

Generalized Convolution Theorem: Given time-domain temporal convolution kernel function $f(t)$ and spatial convolution kernel function $g(t)$, their combination $f(t) * g(t)$ is equivalent to frequency-domain multiplication $F(\omega) \cdot G(\omega)$ as follows:

$$\mathcal{F}\{g * (f * x)\} = G(\omega) \cdot F(\omega) \cdot X(\omega) \quad (18)$$

where $\mathcal{F}$ refers to Fourier Transform operations, x and $X(\omega)$ are the time domain and frequency domain representations of input.

Proof: (1) The convolution of two time-domain signals $f(t)$ and $g(t)$ is defined as:

$$f(t) * g(t) = \int f(\tau)g(t - \tau)d\tau \quad (19)$$

We want to prove that the convolution of $f(t)$ and $g(t)$ in the time domain corresponds to multiplication in the frequency domain.

(2) Calculate the Fourier Transformation of $g * (f * x)$:

$$\begin{aligned}
 & \mathcal{F}\{g * (f * x)\} \\
 &= \int [g * (f * x)] e^{-j\omega t} dt \\
 &= \int \left[\int g(\tau) \cdot (f * x)(t - \tau) d\tau \right] e^{-j\omega t} dt \\
 &= \int g(\tau) \cdot \left[\int (f * x)(t - \tau) \cdot e^{-j\omega t} dt \right] d\tau
 \end{aligned} \tag{20}$$

(3) Let $t' = t - \tau$, then $dt' = dt$, and the above equation becomes:

$$\begin{aligned}
 & \mathcal{F}\{g * (f * x)\} \\
 &= \int g(\tau) \cdot \left[\int (f * x)(t') \cdot e^{-j\omega(t'+\tau)} dt' \right] d\tau \\
 &= \int g(\tau) \cdot \left[\int (f * x)(t') \cdot e^{-j\omega t'} dt' \right] \cdot e^{-j\omega\tau} d\tau \\
 &= \mathcal{F}\{f * x\} * \int g(\tau) e^{-j\omega\tau} d\tau \\
 &= \mathcal{F}\{f * x\} * G(\omega) \\
 &= G(\omega) \cdot F(\omega) \cdot X(\omega)
 \end{aligned} \tag{21}$$

Proved.

C.2 Proof of Discrete Parseval's Theorem

Discrete Parseval's Theorem: For a discrete signal $x(n)$ and its discrete Fourier transformation $X[k]$, the total energy of the signal is equal whether you compute it in the time domain or the frequency domain:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \tag{22}$$

Proof: (1) Using the complex conjugate property $ZZ^* = |Z|^2$, the LHS can be rewritten as:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \sum_{n=0}^{N-1} x[n] x^*[n] \tag{23}$$

(2) Substitute the IDFT formula for $x^*[n]$ and rewrite the LHS from Step 1:

$$\begin{aligned}
 x^*[n] &= \frac{1}{N} \sum_{l=0}^{N-1} X^*[l] e^{-j\frac{2\pi}{N}ln} \\
 \sum_{n=0}^{N-1} x[n] x^*[n] &= \sum_{n=0}^{N-1} x[n] \left(\frac{1}{N} \sum_{l=0}^{N-1} X^*[l] e^{-j\frac{2\pi}{N}ln} \right) \\
 &= \frac{1}{N} \sum_{l=0}^{N-1} X^*[l] \left(\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}ln} \right)
 \end{aligned} \tag{24}$$

(3) Recognize the inner sum as the DFT of $x[n]$:

$$\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}ln} = X[l] \tag{25}$$

(4) Substituting this back into the expression for the LHS and using the complex conjugate property $ZZ^* = |Z|^2$:

$$\begin{aligned}
 \sum_{n=0}^{N-1} x[n] x^*[n] &= \frac{1}{N} \sum_{l=0}^{N-1} X^*[l] X[l] \\
 \sum_{n=0}^{N-1} |x[n]|^2 &= \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2
 \end{aligned} \tag{26}$$

Proved.

D Implementation Details

All experiments are carried out on a Linux workstation equipped with an AMD EPYC 7542 32-Core CPU, an NVIDIA GeForce RTX 4090D GPU, and 512GB of RAM. The code is implemented in Python 3.10 with PyTorch 2.2.2 and CUDA 11.8. To achieve our best practice, we employ AdamW as optimizer with learning rate chosen from $[1e-3, 5e-4, 1e-4]$. The dropout probability is set to 0.2 and weight decay is set to $1e-2$. We also employ Xavier uniform parameter initialization. In addition, we set the number of FFT points to 256, the orthogonal function is set as rFFT normalizer. Our code is available at <https://github.com/JashinKorone/FourierGraphLearning>.

E Complexity Calculation

Firstly, we calculate the theoretical complexity for all baselines: (1) The complexity of RotatE is linear to node number M and edge number N . This is because RotatE is designed for static graphs. When performing rotations in complex space, it only needs to consider the nodes and edges of the current timestep and achieves complexity of $O(M + N)$. (2) For RE-GCN, its complexity is dominated by spatio-temporal convolution and can be calculated by the combination of GNN complexity and RNN complexity:

$$\begin{aligned}
 \Omega_{Re} &= \Omega_{GNN} + \Omega_{RNN} \\
 &= O(TL_g(Md_l + Nd_ld_g)) + O(TN(d_gd_h + d_h^2)) \\
 &\approx O(T(M + N)) + O(TN) \approx O(TM + TN)
 \end{aligned} \tag{27}$$

where the L_g is the layer number of GNNs, each layer has an input feature dimension of d_l and an output feature dimension of d_g , the hidden state of RNN has a dimension of d_h , which corresponds to the features output by the GNN. HisMatch's computational complexity is similar to that of RE-GCN. It includes an additional candidate-related historical structure, so its computational complexity is $O(TM + TN + N)$. (3) SeCoGD is an extension on the basis of RE-GCN, its basic computation cost is similar to RE-GCN. Moreover, it includes an additional hypergraph for cross-context interactions. To this end, the complexity of SeCoGD is $O(TM + TN + TP)$, where P is the average hyperedge number. In practice, P is on the same order of magnitude as M . Therefore, the additional complexity of hypergraph cannot be easily ignored.

Secondly, we analyze the complexity for FGL. Its complexity mainly comes from the fast real Fourier transformation (rFFT) and fast inverse real Fourier transformation (irFFT). Moreover, thanks to the frequency-domain multiplication, the computational complexity for FGNN can be approximated to linear:

$$\begin{aligned}
 \Omega_{FGL} &= 2 \cdot \Omega_{FFT} + \Omega_{mult} \\
 &= 2 \cdot O((TM + N)C_f) + O((TM + N)d_h) \\
 &\approx O(TM + N)
 \end{aligned} \tag{28}$$

Table 7: Results of large-scale dataset benchmark. The best performing model is marked by boldface, the sub-optimal performing model is marked by underline.

Model	ME			EUR		
	MRR	HIT@1	HIT@3	MRR	HIT@1	HIT@3
SeCoGD			<i>Out of Memory</i>			
DHyper	<u>30.29</u>	<u>21.62</u>	<u>33.13</u>	<i>Out of Memory</i>		
DisMult	19.61	10.81	22.68	21.08	13.53	27.43
RotatE	27.05	17.51	29.30	<u>30.56</u>	<u>20.77</u>	<u>35.68</u>
RE-GCN	19.15	11.98	20.40	23.56	16.66	34.67
FGL	38.65	31.23	42.88	42.98	35.73	45.63

where $C_f \ll \min(M, N)$ is the complexity to perform rFFT and irFFT, which is a constant number related to the number of FFT points determined in **Appendix D**. d_h is the embedding size.

F Large-scale Datasets Benchmarks

Beyond the mainstream benchmarks, we incorporate two substantially larger and more diverse ME (Middle East) and EUR (Europe) datasets. This expansion was driven by the observation that real-world social event often transcend beyond one single country, making it necessary for models to predict within broader scope.

The results of large-scale datasets benchmark are shown in Table 7. We highlight two key findings here: (1) All baseline models and FGL performed significantly better than those in mainstream benchmarks. This demonstrates that in social event prediction, limiting the modeling scope to a domestic context may impact performance. Datasets that include all countries within the same region encompass these international interactions, thereby leading to better performance. (2) FGL consistently outperforms other baselines on large-scale datasets. While hypergraph based methods like DHyper and SeCoGD can achieve the sub-optimal performance, their reliance on hyperedges significantly increases memory overhead, often leading to model failure due to Out of Memory (OOM) errors. Although the performance of models like RotatE and DisMult also benefits from larger-scale datasets, a significant gap still remains when compared to FGL.

G Further Discussions

In this section, we discuss the theoretical basis and advantages of FGL and conclude with potential limitations and future works.

G.1 Explanations on Temporal Patterns

In periodic encoder, we utilize a non-trainable sinusoidal cyclical embedding with various frequency patterns to simulate the periodic patterns of social events. The core idea behind this operation is inspired by Fourier series: any periodic or non-periodic signal can be represented as a superposition of a series of sine waves. Based on this insight, in the proposed semantic-aware frequency filter, we utilize the semantics as a guidance to select the most related frequency sections, so as to approximate the real periods of nodes/edges recurrences. The main reason why we don't utilize the recurrences count of each entity and relation is that real-world event data can be noisy and sparse. Directly counting recurrences might be unreliable

if events are infrequent or inconsistently reported. Our methods can interpolate and generalize to utilize temporal patterns even under this circumstance. Moreover, a combination of sine waves can compactly represent correlations and enhance the learning of long-term temporal dependencies.

G.2 Explanations on Fourier Graphs

Note that the long term temporal dependencies between different entities is a very common phenomenon in social event prediction task. For example, the economic instability in some countries can affect the long-term monetary policies of other countries. Moreover, the political turmoil in some countries has led to prolonged refugee crises in other nations. To this end, it is beneficial but challenging to consider long-term dependencies within different entities.

The Fourier Graphs (FGs) address this issue by encoding periodic patterns into nodes and edges, effectively capturing long-term dependencies. Traditional graph-based methods primarily rely on historical time windows to capture both short-term and long-term dependencies. However, due to the gradient vanishing and gradient explosion issues with RNNs, the size of the historical window in these methods cannot be set to a large number. Even worse, existing methods struggle to capture long-term dependencies for events beyond the historical time window. Unlike previous methods, FG incorporates semantic-specific temporal patterns to enable the long-term dependency modeling instead of relying on traditional historical time window, which differs from existing graphs.

G.3 Potential Limitations and Future Works

Although our model can produce competitive results, it does have some limitations: on one hand, FGL utilizes event recurrence as its modeling core, which is compatible with most societal events. However, when faced with extremely rare events or even black swan events, our model lacks reasonable modeling paths. These events often exhibit severe zero-inflated and heavily-tailed characteristics, which can serve as a potential direction for future research. On the other hand, the parameter count for complex-valued modeling is relatively higher, which offers direction for future optimization.

H Ethical Statement

We are committed to advancing social event prediction technology responsibly. The model presented in this research is designed as a decision-support tool intended for positive societal impact, aiding in areas such as disaster response, public safety, and conflict prevention through more accurate and efficient forecasting. All applications of FGL models must strictly adhere to web regulations and internationally recognized ethical standards.

Concurrently, we fully acknowledge the potential risks of misuse and have taken measures accordingly. We recognize that training data (i.e., the GDELT dataset) may contain reporting biases and advise researchers to consider this limitation in application. Moreover, the FGL is designed to be interpretable, with its prediction logic based on transparent periodic patterns. These measures establish a foundation for accountability, serving as a key defense against the model's misinterpretation and abuse.